Nicolás Dalessandro

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Professional Summary

Full-Stack Developer with over 5 years of experience specializing in backend engineering, scalable REST API design, internal tooling, and Angular. Proven track record eliminating mission-critical database bottlenecks (PostgreSQL and Django ORM), developing AI agent integrations via the Model Context Protocol (MCP), and managing containerized Linux infrastructure.

Work Experience

Pabex — Full-Stack Software Developer (Remote)

February 2021 – Present

Client: Cybersecurity & Brand Protection SaaS Platform (March 2023 – Present)

- **Backend Architecture:** Engineered critical REST endpoints (Python and Django REST Framework) for proactive web monitoring, phishing detection, and digital evidence capture modules.
- **Full-Stack Development:** Implemented end-to-end features (Django REST Framework and Angular): case management modules, filtering and record creation, reusable components, and resolution of nested event conflicts.
- **AI Integration (MCP):** Built a self-adaptive Model Context Protocol (MCP) server with custom authentication middleware, enabling autonomous AI agents to dynamically interact with platform tools.
- **API Documentation:** Automated OpenAPI schema generation using drf-spectacular, developing custom extensions to dynamically segment schemas by user role and subscription tier.
- **ORM Optimization:** Refactored critical PostgreSQL queries that stalled tables with timeouts over 5 minutes, slashing latency to milliseconds and restoring system stability.
- **Technical Governance:** Centralized platform architecture into an engineering vault (ADRs) and enforced code review standards in Bitbucket via CODEOWNERS.

Client: Virtual Healthcare Platform (February 2021 – March 2023)

- Developed backend logic in Django for real-time medical appointment scheduling and calendar synchronization.
- Integrated WebRTC peer-to-peer video streaming to power low-latency telehealth consultations directly in the browser.
- Developed the user interfaces for the medical scheduling system using JavaScript and jQuery.

Featured Projects

- **Real-Time Audio Streaming Bot (Node.js, WebSockets, Discord.js):** Distributed system that captures browser audio via a Chrome extension and streams it over WebSockets. Integrated Discord OAuth2 to inject audio into voice channels and deployed via Docker Compose.
- **Cloud Infrastructure (Ubuntu, Nginx, Oracle Cloud):** Administered an ARM-based VPS, Nginx reverse proxy (SSL/TLS), and tuned JVM parameters to optimize performance for a private modded Minecraft server.

Extended technical details of these and other projects at nicolasdalessandro.dev

Technical Skills

- **Backend & APIs:** Python, Django, Django REST Framework, Node.js, OpenAPI, RESTful Architecture.
- **Frontend:** Angular, TypeScript, JavaScript, jQuery, HTML5, CSS3.
- **Databases:** PostgreSQL (Advanced: ORM and Lock Optimization), Redis.
- **DevOps & Infrastructure:** Docker, Docker Compose, Linux (Ubuntu), Nginx, Git, Bitbucket.
- **AI & Integrations:** Model Context Protocol (MCP), WebRTC, WebSockets, Discord API, Mercado Pago.

Education & Languages

- **B.S. in Computer Science:** Universidad Nacional de Río Cuarto (Completed 4 of 5 years).
- **Certifications:** Mercado Pago Checkout Pro (Mercado Libre, Aug 2022).
- **Languages:** Spanish (Native) | English (Intermediate technical reading/writing, basic spoken).